

Gabriel Abreu

ga2740@nyu.edu | (857) 559-3821 | <https://github.com/gmbabreu> | <https://www.linkedin.com/in/gmbabreu/>

EDUCATION

New York University, Courant Institute, New York, NY

Aug 2025 – Present

Doctor of Philosophy (PhD), Mathematics

Brandeis University, Waltham, MA

Aug 2021 – May 2025

BS in Applied Mathematics and Computer Science with Honors

Minor in Economics

GPA: 3.96 | Summa Cum Laude

RESEARCH EXPERIENCE

Applied Math Lab at Courant, New York, NY

May 2026 – Present

Research Assistant — Prof. Leif Ristroph

- Conduct experimental and computational research on real-world fluid dynamics problems, currently investigating the [Feynman sprinkler problem](#)
- Design experiments, develop calibration methodologies, and analyze experimental data using MATLAB.
- Featured in *The New York Times*: [“Physicists Have Argued Over This Problem Since 1883.”](#) describing our research on Feynman sprinkler problem, July 2026

NYU Center for Data Science, New York, NY

Jan 2026 – Present

Research Assistant — Prof. Andrew Gordon Wilson

- Develop efficient and scalable second-order optimization methods for training large language models.
- Implement large-scale training infrastructure in JAX on Cloud TPUs to evaluate Gauss–Newton-style methods.
- Benchmark optimization performance, computational efficiency, and convergence of curvature approximation methods against first-order optimizers such as AdamW.

Maunu Lab, Brandeis University, Waltham, MA

Aug 2024 – May 2025

Research Assistant

- Co-authored manuscript on "A Smoothing Newton Method for Rank-one Matrix Recover" under Prof. Tyler Maunu
- Researched implicit regularization properties of Newton's Method, targeting quadratic convergence for nonsmooth phase retrieval problems
- Developed proofs of stability and investigated phase transitions via the Amplitude Flow algorithm

Data Science Lab, Brandeis University, Waltham, MA

Aug 2023 – May 2025

Intern

- Built an LLM search engine with Retrieval-Augmented Generation (RAG) for university file retrieval, significantly reducing manual processing time
- Deployed an LLM using the AnythingLLM framework in Python for text summarization and analysis across thousands of office files
- Presented findings to the university board and faculty under Prof. Tim Hickey, with results under consideration for adoption across multiple departments; also selected to present at the New England Statistical Society NextGen Data Science Day 2025

Lifespan Lab, Waltham, MA

Jun 2023 – Sep 2024

Lead Undergrad Research Assistant

- Led a team of 6 research assistants to develop machine learning models for automated scoring of audio data from psychology experiments, reducing analysis time and accelerating research progress

- Implemented a model combining IBM Watson and open source code to track repeating sounds in MP3 files, achieving 96% accuracy in automated audio scoring

WORK EXPERIENCE

Brandeis Investment Management, Boston, MA

Jun 2024 – Aug 2024

Analyst Intern

- Evaluated investment decisions and prepared summary reports on green energy and venture capital funds
- Developed an Excel-based carbon model to estimate the total carbon output of investment portfolios, improving sustainability tracking for fund performance

Brandeis Library Scholarly Resources, Waltham, MA

Feb 2022 – Aug 2023

Systems Assistant

- Proposed and implemented an automation program to migrate 1,000+ staff profiles from legacy data archives to a new website
- Processed 3 million XML records in Python to separate them into collections for the library website

LEADERSHIP & INVOLVEMENT

Brazil Conference at Harvard & MIT, Cambridge, MA

Jul 2024 – Present

Vice President

- Oversee the AI4Good program, coordinating the presentation of AI research and applications developed by Brazilian researchers to an international audience at Harvard.
- Mentor Brazilian scholars who have never left the country in developing and presenting open-source AI projects addressing socially relevant problems.
- Support backstage operations for high-profile speakers; guests have included the Fields Medalist Arthur Ávila or former Presidents of Brazil.

TAMID Group, Brandeis University, Waltham, MA

Aug 2022 – Feb 2024

Technology Consultant

- Delivered front-end, back-end, and data analytics solutions for client companies
- Optimized Vcomm's databases by implementing relational data structures, improving storage and retrieval efficiency

Department of Community Living, Brandeis University, Waltham, MA

Aug 2022 – May 2025

Resident Advisor

- Monitored a residence hall, supporting residents with personal concerns and coordinating maintenance resolutions
- Planned and executed community-building programs in collaboration with fellow advisors

Language Empowering Action Project, Brandeis University, Waltham, MA

Jan 2022 – May 2023

Volunteer Tutor

- Taught weekly English classes to adult immigrants and held one-on-one sessions to support individual progress
- Collaborated with a volunteer team to design lesson plans tailored to diverse learning needs

SKILLS

Programming: Python, Java, TensorFlow, Sklearn, Pandas, Matplotlib, spaCy, Flask, Selenium, Git, Jupyter, LaTeX

Software: Excel, VS Code, MS Suite, Eclipse

Languages: English (fluent), Portuguese (native), Spanish (intermediate)